Lab Report No. 8

Digital Signal Processing

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Section: B

Department of Computer Systems Engineering

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CSE 402L: Digital Signal Processing

Demonstration of Concepts	**Poor (Does not meet expectation (1))** The student failed to demonstrate a clear understanding of the assignment concepts	**Fair (Meet Expectation (2-3))** The student demonstrated a clear understanding of some of the assignment concepts	**Good (Exceeds Expectation (4-5)** The student demonstrated a clear understanding of the assignment concepts	**Score** **30%**
Accuracy	The student completed (<50%) tasks and provided MATLAB code and/or Simulink models with errors. Outputs shown are not correct in form of graphs (no labels) and/or tables along with incorrect analysis or remarks.	The student completed partial tasks (50% - <90%) with accurate MATLAB code and/or Simulink models. Correct outputs are shown in form of graphs (without labels) and/or tables along with correct analysis or remarks.	The student completed all required tasks (90%-100%) with accurate MATLAB code and/or Simulink models. Correct outputs are shown in form of labeled graphs and/or tables along with correct analysis or remarks.	**30%**
Following Directions	The student clearly failed to follow the verbal and written instructions to successfully complete the lab	The student failed to follow the some of the verbal and written instructions to successfully complete all requirements of the lab	The student followed the verbal and written instructions to successfully complete requirements of the lab	**20%**
Time Utilization	The student failed to complete even part of the lab in the allotted amount of time	The student failed to complete the entire lab in the allotted amount of time	The student completed the lab in its entirety in the allotted amount of time	**20%**

Lab No: 8

Title Analysis of ASK, FSK, and QAM Modulation & Demodulation using MATLAB

Tasks:

Define Amplitude Shift Keying (ASK).

Amplitude Shift Keying is a digital modulation technique in which the amplitude of a carrier signal is varied according to the digital message signal.

• Define Frequency Shift Keying (FSK).

Frequency Shift Keying (FSK) is a digital modulation technique in which the frequency of the carrier signal is varied according to the digital message signal, while the amplitude and phase of the carrier remain constant

• Define Quadrature Amplitude Modulation (QAM).

Quadrature Amplitude Modulation is a digital modulation technique in which both the amplitude and phase of a carrier signal are varied to represent digital data. It combines two amplitude-modulated signals that are 90° out of phase

• List three applications of digital modulation in communication systems.

1. **Wireless Communication:** Digital modulation techniques such as QAM, PSK, and FSK are used in cellular networks (4G/LTE, 5G) and Wi-Fi to transmit data over radio channels with high spectral efficiency and noise immunity.
2. **Satellite and Cable Communication:** Digital modulation enables transmission of voice, video, and data over long distances. For example, Direct-to-Home (DTH) satellite TV uses QPSK or 16-QAM, while cable broadband systems use 64-QAM or 256-QAM.
3. **Data Transmission over Wired and Optical Channels:** Digital modulation encodes binary data for transmission over fiber optic networks, DSL lines, or power lines, allowing high-speed, reliable data transfer.

Task 1:

• Create and plot a binary message signal in MATLAB.

Hint:

```
data = randi([0 1], 1000, 1);
```

```
% --- TASK 1: CREATE AND PLOT A BINARY MESSAGE SIGNAL ---
clc; clear; close all;
% Clear command window, workspace, and close figures

% Generate 1000 random bits (0 or 1)
data = randi([0 1], 1000, 1);

% --- Parameter Setup (Fixes Nyquist Error) ---
fs_data_rate = 1000;              % R_b: Original Data Rate (Bits/sec)
samples_per_symbol = 20;          % Sp: Samples used to represent each symbol/bit
% Fs: Waveform Sampling Frequency (20000 Hz). High enough to satisfy fskmod.
fs_waveform = fs_data_rate * samples_per_symbol;

% Total time
T = length(data) / fs_data_rate;

% High-resolution time vector (Used for FSK and plotting ASK/QAM waveforms)
t = linspace(0, T, T*fs_waveform);
% Time vector for plotting the original bits
t_data = linspace(0, T, length(data));
```

```

% Plot the Binary Message Signal
figure;
stairs(t_data, data, 'LineWidth', 1.5);
ylim([-0.2 1.2]);
xlabel('Time (s)');
ylabel('Amplitude');
title('Binary Message Signal');
grid on;
```

Figure 1: Code for the binary message signal

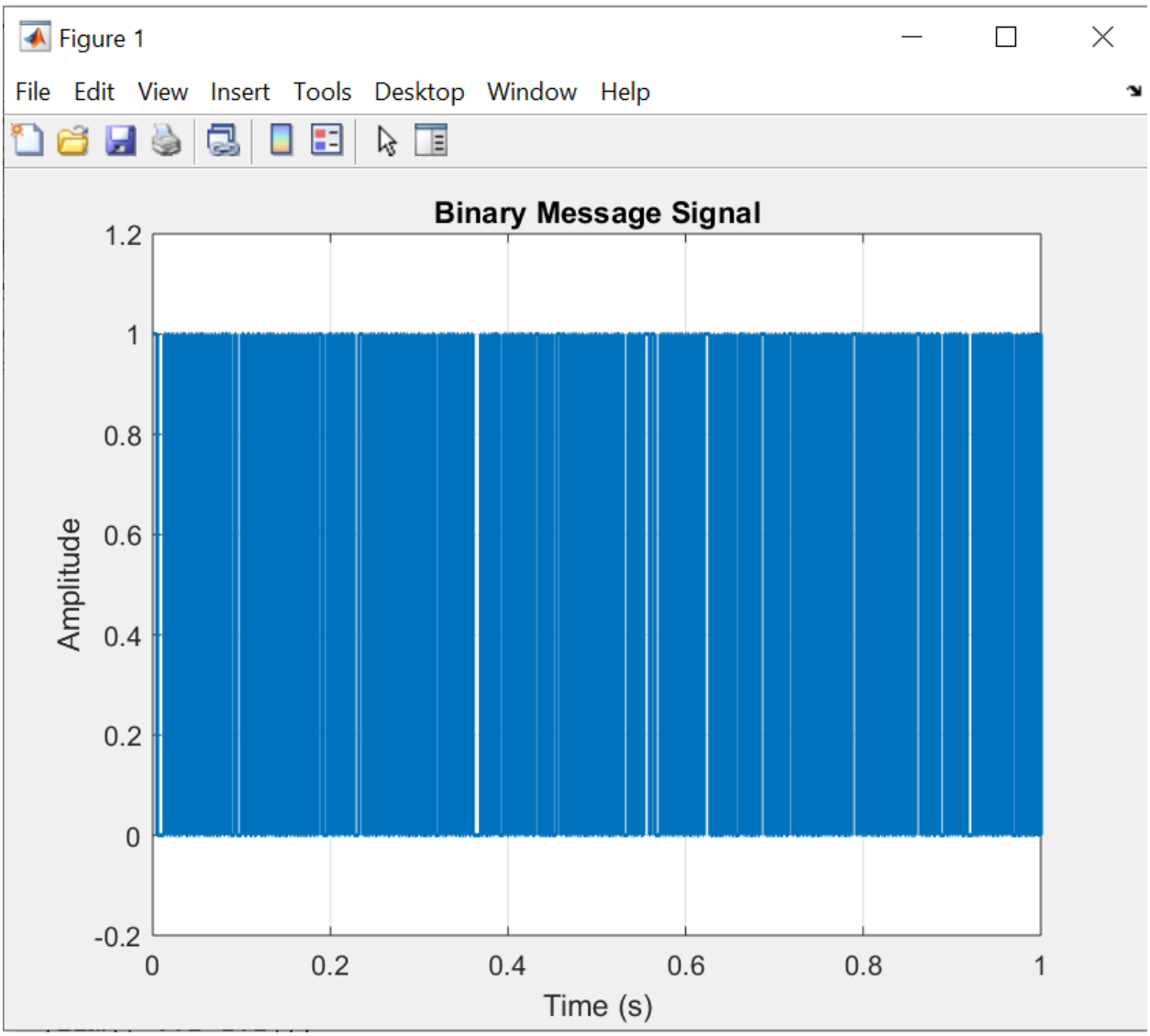


Figure 2:Plot of the binary signal message

Task 2:

• Generate carrier signals for ASK, FSK, and QAM.

```
% --- TASK 2: GENERATE CARRIER SIGNALS ---
fc_ask = 50;
fc_qam = 60;

% Carrier signals generated for the high-resolution time vector 't'
carrier_ask = sin(2*pi*fc_ask*t);
carrier_i_qam = cos(2*pi*fc_qam*t); % In-phase (I) carrier
carrier_q_qam = sin(2*pi*fc_qam*t); % Quadrature (Q) carrier
```

Command Window

Figure 3Code for the ASK,FSK and QAM

Task 3:

• Modulate the signal using built-in MATLAB functions: askmod, fskmod, qammod.

Hint:

M = 2;
ask_mod = askmod(data, M);

freq_sep = 5;
fsk_mod = fskmod(data, M, freq_sep, 100);

M = 16;
qam_mod = qammod(data, M, 'InputType', 'bit');

```
% --- TASK 3 & 4: MODULATION AND DEMODULATION ---

% 1. ASK Modulation & Demodulation (M=2 is Binary ASK/PAM)
M_ask = 2;
% R2019a syntax: Modulate in the symbol domain (1 output value per bit)
ask_mod_symbol = pammod(data, M_ask);
ask_demod = pamdemod(ask_mod_symbol, M_ask);

% 2. FSK Modulation & Demodulation
M_fsk = 2;
freq_sep = 5;
```

```
% FSKmod/demod syntax requires all 5 arguments to output the waveform
% and satisfy the Nyquist condition (Fs > 2*max_freq)
fsk_mod = fskmod(data, M_fsk, freq_sep, fs_waveform, samples_per_symbol);
fsk_demod = fskdemod(fsk_mod, M_fsk, freq_sep, fs_waveform, samples_per_symbol);

% 3. QAM Modulation & Demodulation (M=16)
M_qam = 16;
% R2019a syntax: Modulate in the symbol domain
qam_mod_symbol = qammod(data, M_qam, 'InputType', 'bit');
qam_demod = qamdemod(qam_mod_symbol, M_qam, 'OutputType', 'bit');


% --- MANUAL WAVEFORM GENERATION (Needed for ASK/QAM plotting) ---
% Since pammod/qammod do not output the full waveform, we manually upsample
% the symbol-domain output to create the high-resolution signal for plotting.
ask_mod = reshape(repmat(ask_mod_symbol.', samples_per_symbol, 1), 1, []);
qam_mod = reshape(repmat(qam_mod_symbol.', samples_per_symbol, 1), 1, []);
qam_mod = qam_mod.'; % Ensure it is a column vector
```

Task 4:

• Demodulate using askdemod, fskdemod, qamdemod.

Task 5:

• Plot time and frequency domain representations.

figure; subplot(2,1,1); stem(ask_mod); title('ASK Modulated');
subplot(2,1,2); stem(ask_demod); title('ASK Demodulated');

figure; subplot(2,1,1); plot(fsk_mod); title('FSK Modulated');
subplot(2,1,2); plot(fsk_demod); title('FSK Demodulated');

```
% --- TASK 5: PLOT TIME DOMAIN REPRESENTATIONS ---

% 1. ASK Plot
figure;
subplot(2,1,1);
plot(t(1:400), ask_mod(1:400));
title('ASK Modulated Waveform (First 20 Bits)');
xlabel('Time (s)'); ylabel('Amplitude');

subplot(2,1,2);
stem(ask_demod(1:20)); % Demodulated signal (20 bits)
title('ASK Demodulated Bits (First 20)');
xlabel('Bit Index'); ylabel('Bit Value');
```

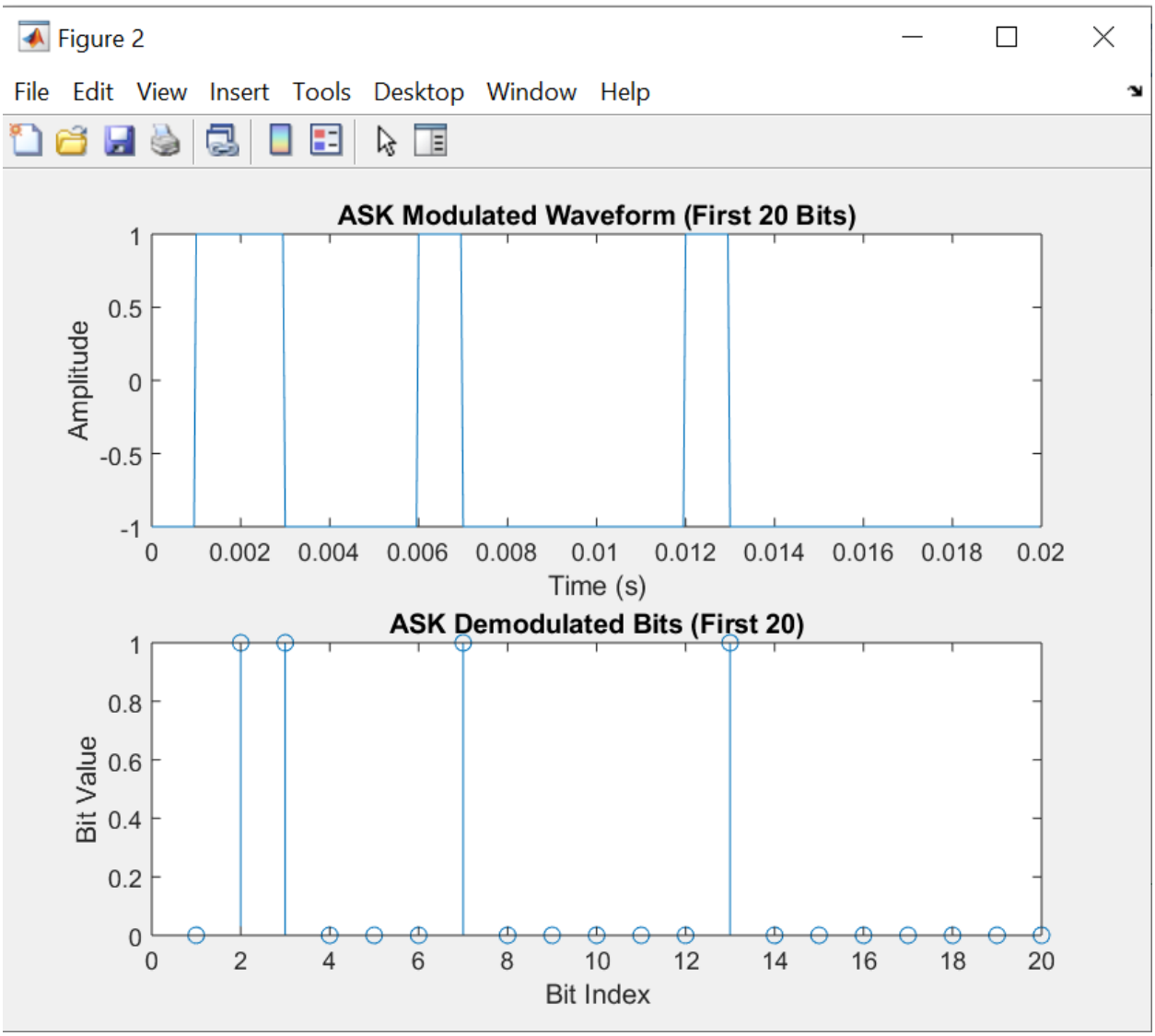
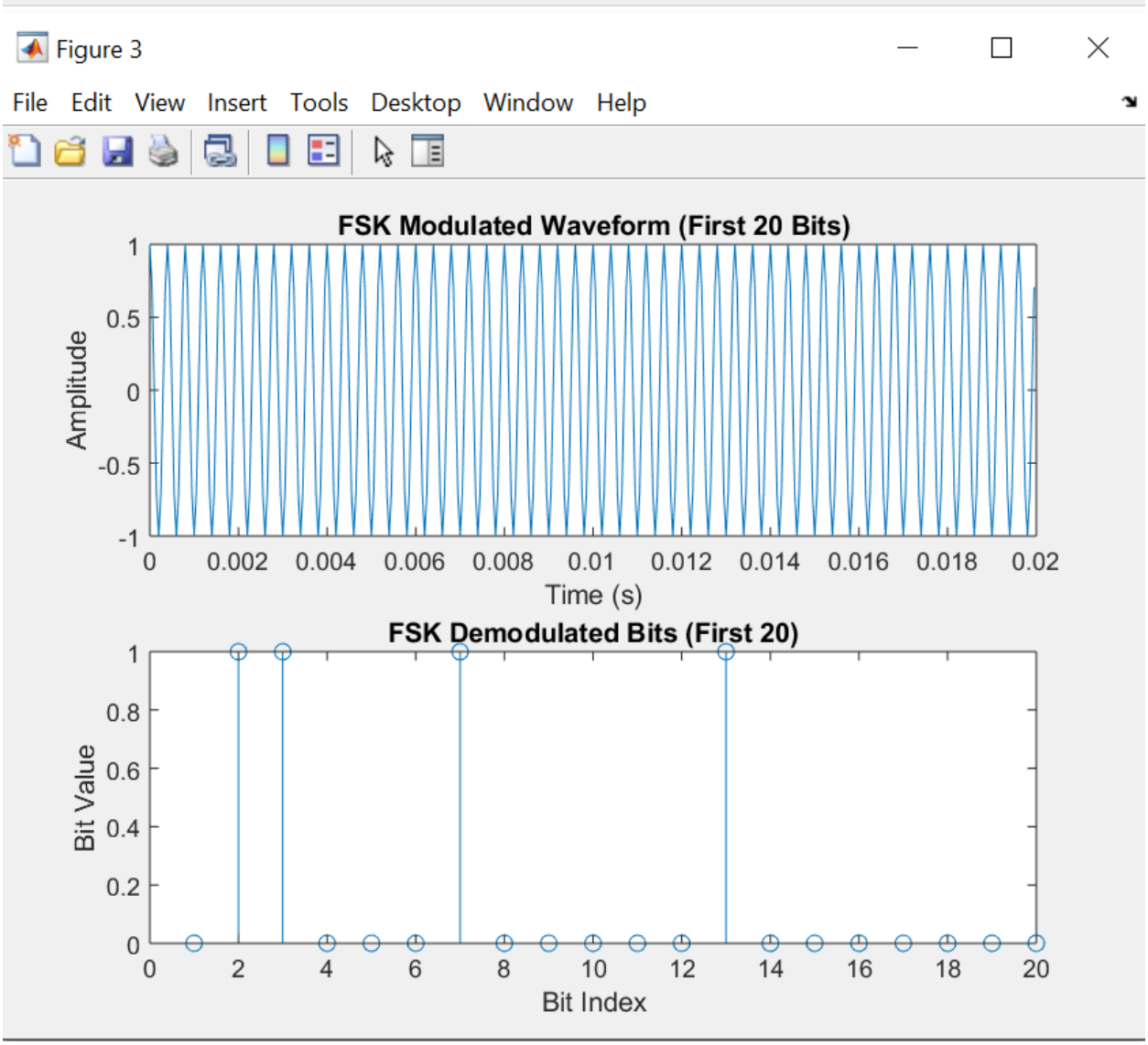


Figure 2
File Edit View Insert Tools Desktop Window Help
ASK Modulated Waveform (First 20 Bits)
Amplitude
1
0.5
0
-0.5
-1
0 0.002 0.004 0.006 0.008 0.01 0.012 0.014 0.016 0.018 0.02
Time (s)
ASK Demodulated Bits (First 20)
Bit Value
1
0.8
0.6
0.4
0.2
0
0 2 4 6 8 10 12 14 16 18 20
Bit Index

```
% 2. FSK Plot
figure;
subplot(2,1,1);
% Plot the FSK modulated waveform (frequency changes when the bit changes)
plot(t(1:400), real(fsk_mod(1:400)));
title('FSK Modulated Waveform (First 20 Bits)');
xlabel('Time (s)'); ylabel('Amplitude');

subplot(2,1,2);
stem(fsk_demod(1:20)); % Demodulated signal (20 bits)
title('FSK Demodulated Bits (First 20)');
xlabel('Bit Index'); ylabel('Bit Value');
```



Task 6:

• Analyze the modulation and demodulation results.

figure; subplot(2,1,1); scatter(real(qam_mod), imag(qam_mod)); title('QAM Constellation');
subplot(2,1,2); stem(qam_demod); title('QAM Demodulated');

```
% --- TASK 6: ANALYZE THE MODULATION AND DEMODULATION RESULTS ---

% 3. QAM Constellation and Demodulated Plot
figure;
subplot(2,1,1);
% The QAM constellation diagram plots the symbol-domain signal (I vs Q)
scatter(real(qam_mod_symbol), imag(qam_mod_symbol), 'filled');
title('QAM Constellation (M=16)');
xlabel('In-phase'); ylabel('Quadrature');
grid on;

subplot(2,1,2);
% Demodulated signal (20 bits)
stem(qam_demod(1:20));
title('QAM Demodulated Bits (First 20)');
xlabel('Bit Index');
ylabel('Bit Value');
grid on;
```

Command Window

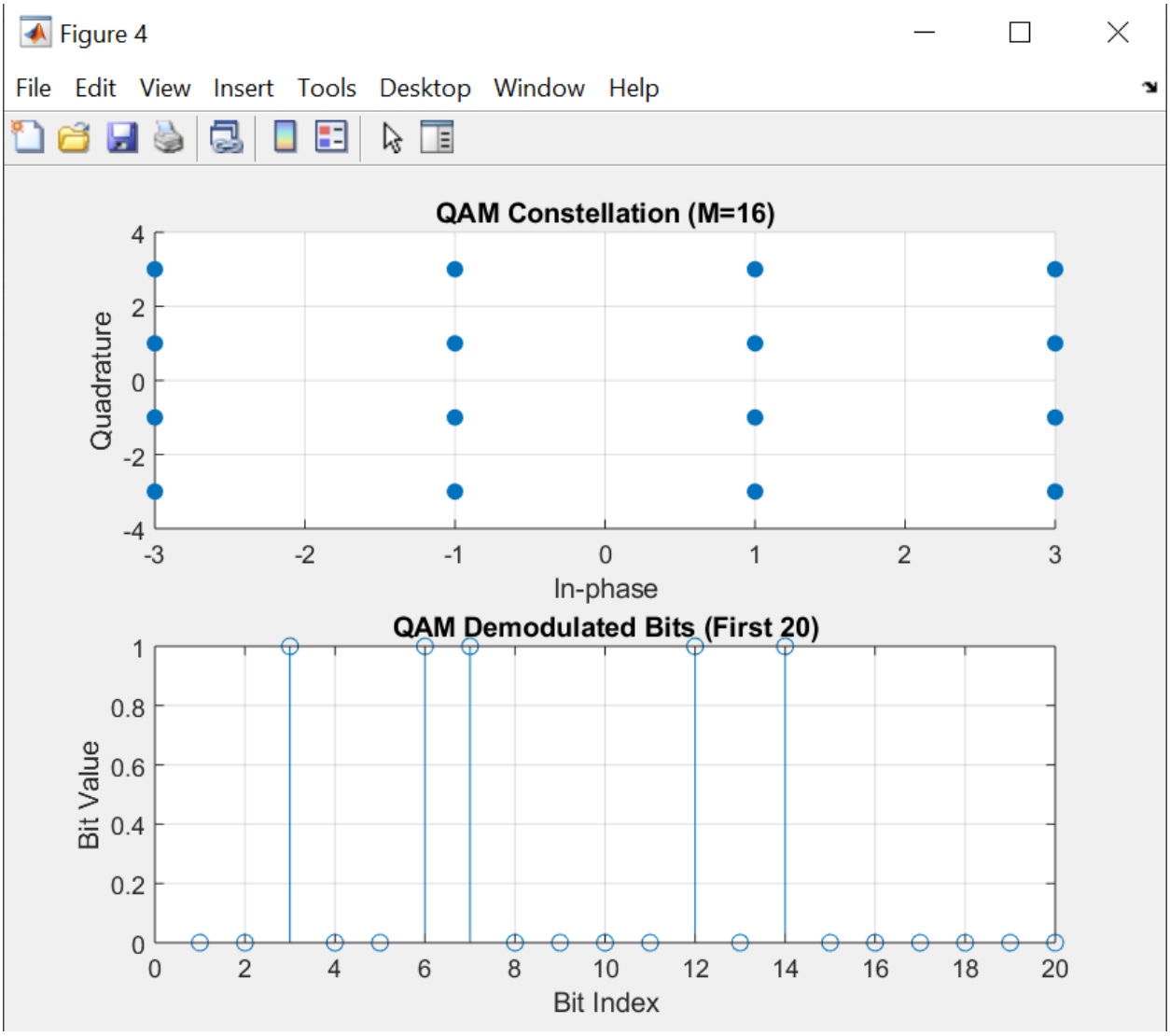


Figure 4
File Edit View Insert Tools Desktop Window Help
QAM Constellation (M=16)
Quadrature
In-phase
4
2
0
-2
-4
-3
-2
-1
0
1
2
3
QAM Demodulated Bits (First 20)
Bit Value
Bit Index
1
0.8
0.6
0.4
0.2
0
0
2
4
6
8
10
12
14
16
18
20